

JABER VALINEJAD

Boston, MA · Jaber.Valinejad@Gmail.Com · (571) 489-2230 · LinkedIn · GitHub

Immigration status: Permanent resident (Green Card holder)

PROFILE SUMMARY

Generative AI scientist with 10+ years of experience in building advanced solutions across large language models (LLMs), data science, machine learning, and NLP. Proven leader in developing and deploying scalable AI systems in biomedical, social, and infrastructure domains. Career highlights include:

- Architected enterprise-level LLM pipelines integrating Retrieval-Augmented Generation (RAG), LoRA/PEFT fine-tuning, and AI agents for large-scale search, recommendation, and automation use cases.
- Built and deployed robust AI applications using LLaMA, BioMedBERT, GPT-4, and OpenAI APIs, integrating REST endpoints for real-time LLM operations.
- Track record of leading GenAI projects involving RAG pipelines, LoRA/PEFT fine-tuning, agent-based reasoning, and knowledge graph modeling using LLaMA, BioMedBERT, and GPT models.
- Skilled in deploying multi-modal LLM solutions on AWS, Azure, and GCP, with expertise in LLMOps and scalable MLOps best practices.
- Experienced in public speaking, grant writing, and presenting cutting-edge research at international conferences and technical forums.
- Adept at working in cross-functional teams, collaborating with academic, industry, and government stakeholders to translate AI research into impactful applications.
- Passionate about applying GenAI to real-world challenges.

EXPERIENCE

National Center for Advancing Translational Sciences, NIH

Remote

Generative AI Engineer

October 2023-Present

- o Built a large-scale LLM-powered social knowledge graph (3M+ researchers) using LangGraph, RAG, and Neo4j to recommend rare disease specialists via semantic clustering and expertise summarization.
- o Fine-tuned BioMedBERT, LLaMA-3, and GPT models for rare disease detection across 320K+ NIH projects, integrating outputs into a searchable knowledge graph mapped to 2,000+ diseases.
- o Developed a drug discovery pipeline combining LLM fine-tuning (TinyLlama), phenotypic screening, and semantic pathway analysis using AWS EC2/S3, PubChem, CTD, and Neo4j.
- o Engineered a cloud-native ETL + knowledge graph framework (ADE4RD) for pharmacovigilance, processing FAERS data with parallel pipelines and Dockerized Neo4j deployment.
- o Deployed scalable GenAI systems across AWS and Azure with LLMOps best practices, using SageMaker, LoRA/PEFT, and prompt engineering (CoT, self-feedback) to optimize biomedical AI workflows.

Data and System Science in Public Health Lab, Harvard University

Boston, MA

Healthcare Data Scientist

February 2022-September 2023

- o Analyzed 1TB+ NEMESIS and 20B+ MarketScan records to model disease progression, EMS response, and treatment disparities using Azure Databricks (PySpark), AWS EC2, and SQL.
- o Built scalable pipelines for longitudinal modeling, cohort analysis, and risk prediction, identifying outcome drivers across demographics, insurance types, and geographic regions.
- o Developed dashboards in Tableau and conducted spatiotemporal analysis to visualize healthcare equity gaps and inform data-driven public health decisions.

Disaster Resilience and Risk Management Research Center

Falls Church, VA

AI Research Scientist – NLP & Risk Intelligence

January 2020- December 2021

- o Applied NLP, machine learning, and social computing to analyze public discourse and extract behavioral and resilience indicators from text data related to urban flooding.
- o Developed data-driven risk models combining structured and unstructured data to assess healthcare and infrastructure resilience, including DEA-based evaluation of emergency services.
- o Collaborated across multi-disciplinary teams to build scalable, explainable frameworks for community resilience modeling, setting the foundation for early GenAI applications in disaster informatics.

Shanghai Center for Artificial Intelligence and Data Analytics

Falls Church, VA

NLP Engineer – Social Analytics & AI

January 2020-December 2021

- o Applied deep learning (CNN, RNN, LSTM, GANs) and NLP techniques to evaluate social-psychological resilience across five countries using 43M tweets and 5K news articles during the COVID-19 pandemic.
- o Built multilingual, multi-modal pipelines for fake news detection, sentiment analysis, and community well-being scoring, contributing to three peer-reviewed publications.
- o Collaborated on five cross-cultural projects with international teams, delivering data-driven insights to support public policy and crisis management.

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Departments of ECE and Computer Science, Virginia Tech

Research Data Associate

Falls Church, VA

August 2018- December 2021

- o Developed a multi-agent AI system integrating NLP-based social sensing, infrastructure data (FEMA, utilities), and behavioral modeling to simulate and quantify disaster resilience across 3,325 U.S. counties.
- o Applied psychological text mining (LIWC, SEANCE) to analyze 490K+ tweets and media for emotion, misinformation, and adaptability; validated models with $R^2 > 0.9$ using real-world disaster timelines.
- o Authored 10 research publications, delivered 6 international presentations, supported \$500K in grant proposals, and contributed to multiple projects across data science, N-DEA, and resilience forecasting.

Mazandaran Regional Electric Company (MAZREC)[†]

Energy Data Specialist

Sary, Iran

November 2013-August 2018

- o Led the development of SmartGridGuard, a real-time anomaly detection and simulation platform for high-voltage transmission systems under environmental stress, integrating pollution and weather data into insulation coordination models.
- o Built a modular diagnostic pipeline using MATLAB, C++, Python, and SQL, reducing outage duration by 8%, preventing 130+ annual failures, and saving over 200 MWh of energy in high-risk regions.
- o Collected and harmonized data from 10+ external sources including power utilities, meteorological agencies, and equipment manufacturers to drive simulation accuracy and predictive analytics.

SKILLS

- Generative AI: RLHF, Multi-Modal language models, Diffusion models
- Data Analysis: Python, Pandas, NumPy, Scikit-learn, Git, C/C++ and R
- Deep Learning Frameworks: PyTorch, TensorFlow, JAX
- Database Management: SQL, Neo4j
- Cloud Technologies: AWS (S3, EC2, SageMaker), Azure, GCP
- Reinforcement learning: Gymnasium, stable_baselines
- Containerization: Docker
- Statistical Methods: A/B experiment, Regression, Classification, Dimension Reduction, Clustering, Causal Inference
- Data Visualization: Tableau, Matplotlib, Seaborn
- UX/UI design: Figma, Gradio
- Technical Writing and Public Speaking

EDUCATION

Harvard University, Postdoctoral Fellow

Data and System Science in Public Health Lab, Boston, MA

Virginia Tech, PhD[†]

Electrical and Computer Engineering, Falls Church, VA

Virginia Tech, MSc[†]

Computer Science, Falls Church, VA

Noshirvani University of Technology, BS

Electrical and Computer Engineering, Babol, Iran

CERTIFICATES

- Advanced Deep Learning: Large-Scale Distributed Training (JAX)
- Reinforcement Learning: Reward Modeling for LLMs
- Fine-Tuning LLaMA & GPT with HuggingFace
- LangGraph: AI Agents for RAG and Automation Workflows